EE 332 Introduction to IOT
Project Report

Title: IoT-Based Health Monitoring and Medication Adherence System

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Abstract : This project presents the design and implementation of an IoT-based health monitoring and medication adherence system aimed at assisting patients, particularly elderly individuals, in maintaining proper medication schedules and monitoring vital health parameters. In many real-life situations, patients tend to forget or miss their prescribed medicines, which can lead to serious health complications. To address this issue, the proposed system integrates medication reminder functionality with real-time health monitoring into a single platform. The system is built around the ESP32 microcontroller, which enables wireless communication using WiFi. It is interfaced with sensors such as the MAX30102 for measuring heart rate and SpO_2 levels, and the DHT11 sensor for temperature monitoring, as a dedicated body temperature sensor was not available. A display unit is used to present patient information, reminders and sensor readings. A mobile application developed using Flutter allows doctors to enter prescriptions, including medicine names and timings. This data is stored in Firebase Realtime Database and accessed by the ESP32. The system compares the current time with stored schedules to trigger reminders using a buzzer and LED-based tray indication, ensuring the correct medicine is taken. Additionally, the system logs whether medicines are taken or missed and uploads health data to Firebase for remote monitoring. An emergency alert feature is also included for critical situations. Overall, the system provides a reliable and efficient IoT-based solution that improves medication adherence and enables real-time health monitoring.

Keywords : IoT, ESP32, Firebase Realtime Database, Flutter, HTTP/HTTPS, WiFi Communication, I2C Protocol, SPI Protocol, NTP Time Synchronization, MAX30102, DHT11, Medication Reminder System, Remote Health Monitoring, Emergency Alert System

Introduction :

In recent years, the application of Internet of Things (IoT) technology in healthcare systems has gained significant attention due to its ability to provide continuous monitoring, remote accessibility, and timely medical assistance. With the increasing aging population and prevalence of chronic diseases, ensuring proper medication adherence and regular health monitoring has become a critical challenge. IoT-based systems address this issue by enabling communication between sensors, embedded devices and cloud platforms. Existing healthcare solutions are generally classified into health monitoring systems and medication reminder systems. Health monitoring systems focus on measuring physiological parameters such as heart rate, temperature, and blood oxygen levels, but often lack mechanisms to ensure adherence to prescribed medication schedules. On the other hand, traditional medication reminder systems, including alarm-based devices, provide basic alerts without offering monitoring, tracking, or remote supervision. Recent mobile application-based solutions improve usability; however, many lack real-time synchronization with hardware devices and do not provide an integrated approach combining monitoring and adherence tracking. To overcome these limitations, this work proposes an integrated IoT-based system that combines medication reminders with real-time health monitoring.The proposed system utilizes an ESP32 microcontroller interfaced with biomedical sensors for data acquisition, while Firebase Realtime Database enables cloud-based synchronization with a Flutter-based mobile application. This integrated approach enhances system reliability, supports remote monitoring, and provides a practical solution for modern healthcare applications.

Motivation behind the project :

The motivation for this project comes from my village background, where I have observed that many elderly people are often left alone at home while other family members go to work in the fields or outside. In such situations, elderly people may face difficulty in asking for help during emergencies, especially if they are not comfortable using smartphones or modern devices. They may also forget to take medicines on time or may take the wrong medicine by mistake. These problems can affect their health and safety, particularly in rural areas where immediate medical support may not always be available. To address these challenges, this project proposes a simple IoT-based system that provides medicine reminders, basic health monitoring, and an emergency alert feature using a single button press. The main aim of this project is to improve the safety, healthcare access, and independence of elderly people, especially in rural environments.

Objectives :

The main objectives of this project are as follows:

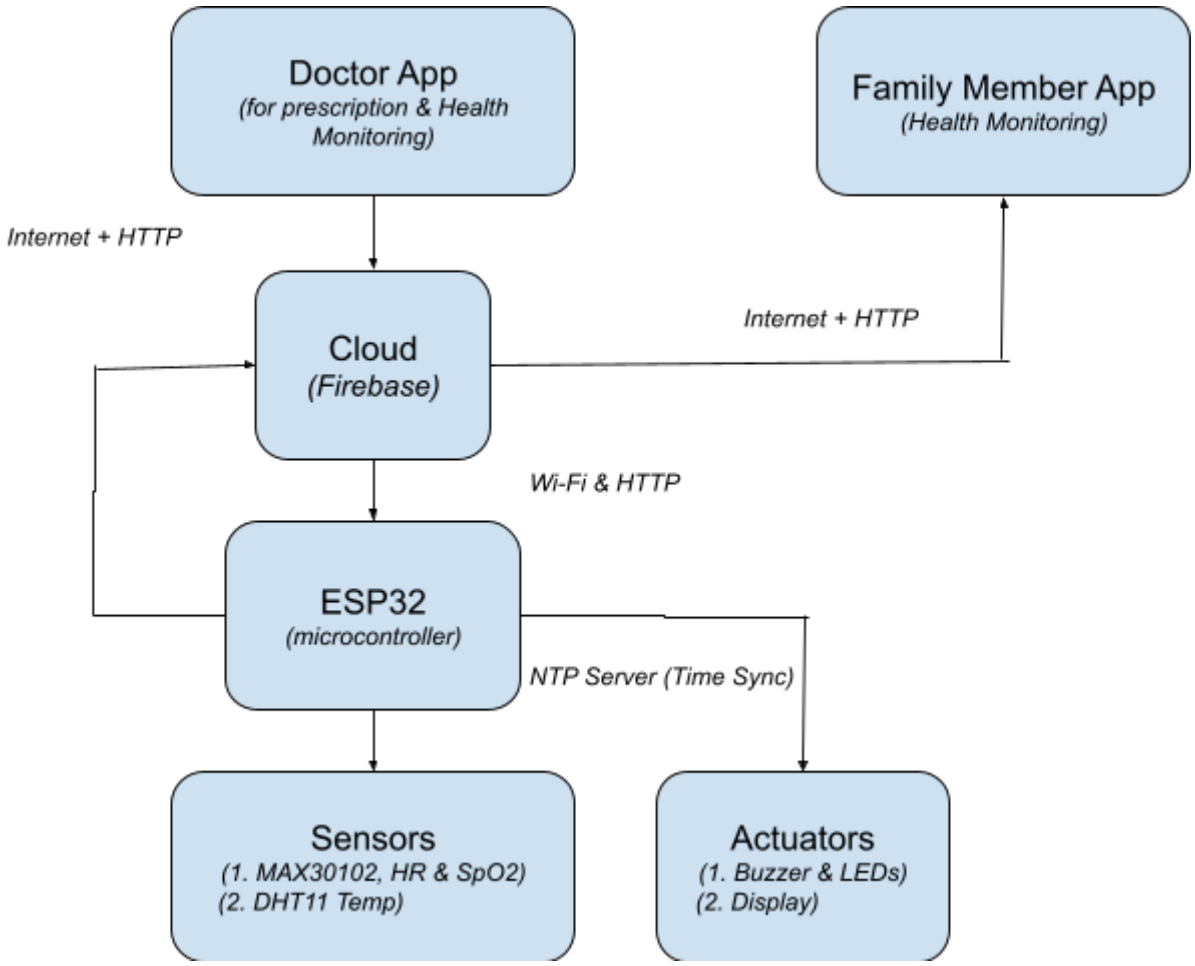
- To develop an IoT-based system for providing timely medication reminders.
- To monitor vital health parameters such as heart rate, SpO_2, and body temperature.
- To track medication adherence by recording whether medicines are taken or missed.
- To enable remote monitoring of patient data through a mobile application.
- To implement an emergency alert mechanism for immediate assistance.

Methodology :

The proposed system is designed based on an Internet of Things (IoT) architecture that integrates embedded hardware, cloud services, and a mobile application to provide medication reminders and health monitoring. The ESP32 microcontroller is used as the core processing unit due to its built-in WiFi capability, enabling seamless communication with the cloud. The system incorporates biomedical sensors such as the MAX30102 for measuring heart rate and SpO_2, and the DHT11 sensor for monitoring temperature. These sensors continuously acquire physiological data from the user. A mobile application is developed using Flutter to provide an interface for entering prescription details, including medicine names, timings, and instructions. The data is stored in Firebase Realtime Database, which acts as the cloud backend and ensures real-time synchronization between the device and the application. Communication between the ESP32 and Firebase is established using the HTTP protocol over WiFi. HTTP is selected due to its simplicity, wide support, and compatibility with Firebase services. It enables reliable data transmission for uploading sensor readings, retrieving prescription data, and updating logs. Since the system operates with periodic data exchange rather than continuous streaming, HTTP provides an efficient and practical solution without requiring additional infrastructure such as message brokers. The ESP32 periodically retrieves prescription data from Firebase and compares it with the current time obtained using Network Time Protocol (NTP). When the scheduled time matches, the system generates an alert using a buzzer and LED-based tray indication, guiding the user to the appropriate medicine. Furthermore, the system records medication adherence by detecting user input through buttons indicating whether the medicine is taken or missed. The collected health data and adherence logs are uploaded to Firebase, allowing remote monitoring through the mobile application. An emergency alert mechanism is also implemented, where a dedicated button triggers

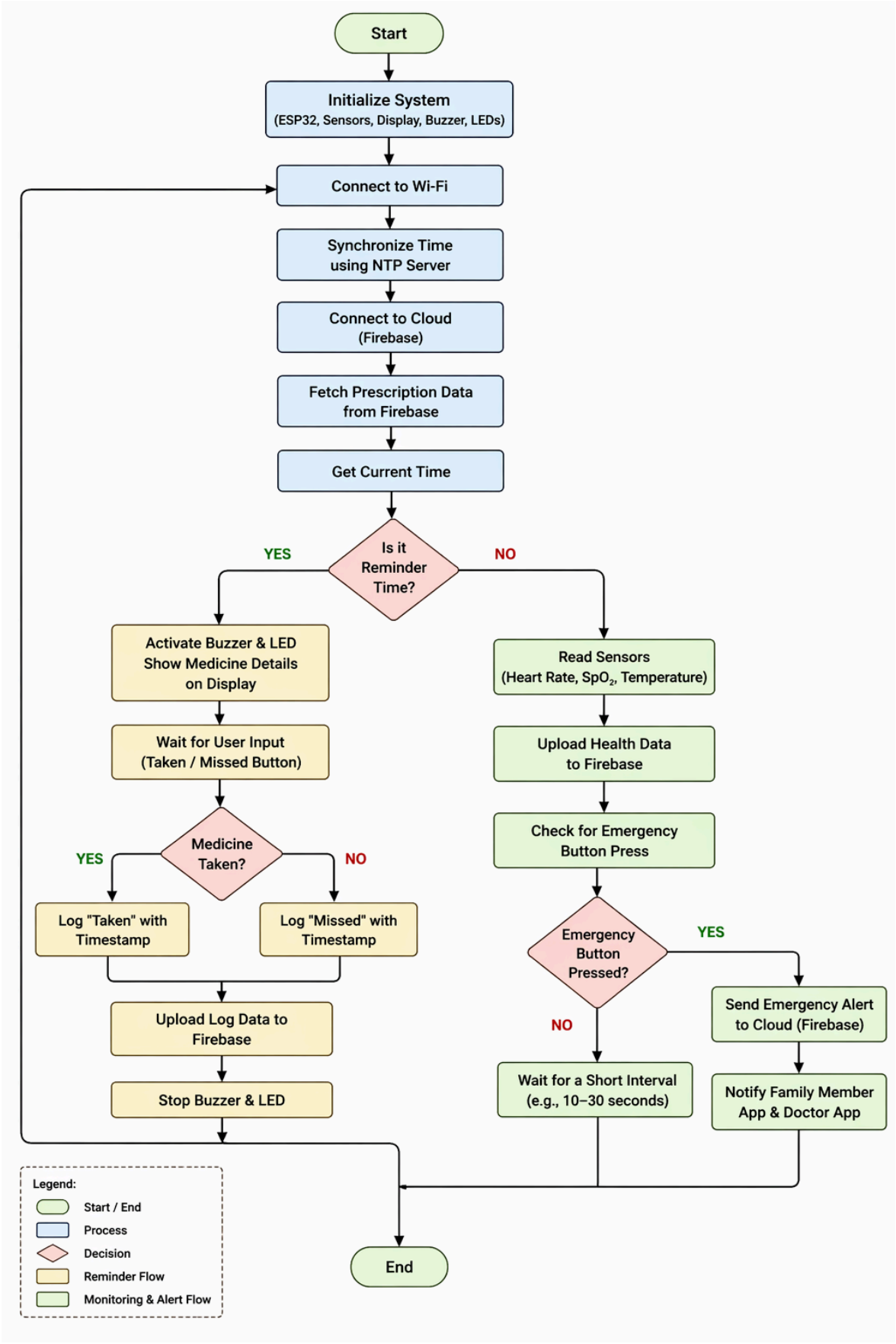
an alert message that is sent to the cloud, enabling immediate response from family members or healthcare providers. Overall, the proposed design provides a simple, reliable and scalable IoT-based solution for improving medication adherence and enabling real-time health monitoring.

Block Diagram :



Flowchart :

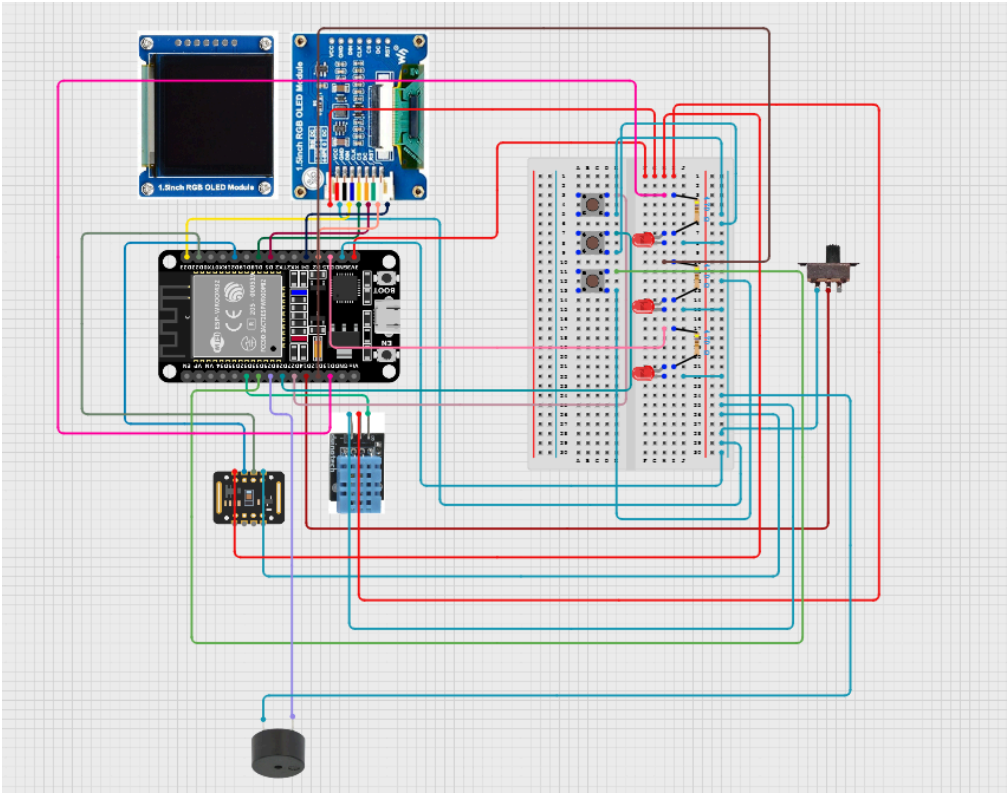
The system follows a sequential flow starting from initialization, data synchronization, reminder checking, and sensor monitoring. The ESP32 continuously checks for scheduled reminders and user inputs while updating data to the cloud.


Start
Initialize System
(ESP32, Sensors, Display, Buzzer, LEDs)
Connect to Wi-Fi
Synchronize Time
using NTP Server
Connect to Cloud
(Firebase)
Fetch Prescription Data
from Firebase
Get Current Time
Is it
Reminder
Time?
YES
NO
Activate Buzzer & LED
Show Medicine Details
on Display
Wait for User Input
(Taken / Missed Button)
Medicine
Taken?
YES
NO
Log "Taken" with
Timestamp
Log "Missed" with
Timestamp
Upload Log Data to
Firebase
Stop Buzzer & LED
Read Sensors
(Heart Rate, SpO2, Temperature)
Upload Health Data
to Firebase
Check for Emergency
Button Press
Emergency
Button
Pressed?
YES
NO
Send Emergency Alert
to Cloud (Firebase)
Wait for a Short Interval
(e.g., 10–30 seconds)
Notify Family Member
App & Doctor App
End
Legend:
Start / End
Process
Decision
Reminder Flow
Monitoring & Alert Flow

Components used :

- **ESP32:** Main microcontroller used for processing and WiFi communication with Firebase (HTTP over WiFi); controls sensors, display, buzzer and LEDs.
- **MAX30102:** Used for heart rate and SpO_2 measurement; connected via **I2C (SDA, SCL pins)** to ESP32.
- **DHT11:** Used for temperature measurement; connected to a **digital GPIO pin** of ESP32.
- **OLED Display (SSD1351):** Displays patient data and reminders; connected via **SPI (MOSI, SCK, CS, DC, RST pins)**.
- **Buzzer:** Provides sound alert during reminders; connected to a **digital output pin (e.g., GPIO 25)**.
- **LEDs (Tray Indicators):** Indicate medicine trays; connected to **GPIO pins (e.g., GPIO 12, 13, 15)**.
- **Push Buttons:** Used for Taken, Missed, and Emergency inputs; connected to **GPIO pins with INPUT_PULLUP configuration**.
- **Firebase Realtime Database:** Cloud backend used for storing prescriptions, health data, and logs; communication via **HTTP protocol over WiFi**.
- **Mobile Application (Flutter):** Used by doctor and family members for prescription entry and monitoring; communicates with Firebase via **Internet (HTTP/HTTPS)**.
- **WiFi:** Communication medium enabling ESP32 to connect to cloud services for real-time data exchange.

Pin connections :



Display Model - OLED Display SSD1351

OLED to ESP32 Pin Mapping

OLED Pin	ESP32 Pin
VCC	3.3V
GND	GND
DIN / MOSI	GPIO 23
CLK / SCK	GPIO 18
CS	GPIO 5
DC	GPIO 2
RST	GPIO 4

MAX30102 to ESP32 Pin Mapping

MAX30102 Pin	ESP32 Pin
VIN / VCC	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

DHT11 to ESP32 Pin Mapping

DHT11 Pin	ESP32 Pin
S / DATA	GPIO 32
+ / VCC	3.3V
- / GND	GND

Button to ESP32 Pin Mapping

Button	ESP32 Pin
TAKEN Button	GPIO 27
MISSED Button	GPIO 26
Emergency Button	GPIO 33

Tray LED to ESP32 Pin Mapping

Tray LED	ESP32 Pin
Tray 1 LED	GPIO 13
Tray 2 LED	GPIO 12
Tray 3 LED	GPIO 15

Buzzer to ESP32 Pin Mapping

Buzzer Pin	ESP32 Pin
+	GPIO 25
-	GND

Slide Switch to ESP32 Connection

Slide Switch Pin	ESP32 Connection
Middle pin / COM	GPIO 14
One side pin	GND
Other side pin	Not connected

Code Developed :

ESP32 Code Logic :

The embedded software for the proposed system is developed using the Arduino IDE for the ESP32 microcontroller. The program is structured to handle multiple functionalities, including WiFi communication, cloud data exchange, sensor data acquisition, reminder generation, and user interaction.

Several libraries are utilized to simplify development and ensure reliable operation. The WiFi.h library is used to establish internet connectivity, while Firebase_ESP_Client.h enables communication with the Firebase Realtime Database using the HTTP/HTTPS protocol. Time synchronization is achieved using the time.h library through Network Time Protocol (NTP), which is essential for accurate reminder triggering. For peripheral interfacing, the SPI.h library is used for communication with the OLED display, along with Adafruit_GFX.h and Adafruit_SSD1351.h for graphics and display control. The Wire.h library is used for I2C communication with the MAX30102 sensor, which is supported by MAX30105.h and associated algorithms for calculating heart rate and SpO_2 values. The DHT.h library is used for reading temperature data from the DHT11 sensor. The system uses multiple communication protocols.

WiFi is used as the communication medium, while HTTP/HTTPS is used for data exchange with Firebase. The OLED display communicates using the SPI protocol, and the MAX30102 sensor uses the I2C protocol. Digital GPIO pins are used to interface push buttons, buzzers, LEDs, and the slide switch.

The program execution begins with system initialization, where all peripherals and connections are configured. The ESP32 first connects to WiFi, synchronizes time using NTP, and initializes Firebase. It then retrieves prescription data from Firebase, including medicine details, timings, and tray assignments. During operation, the ESP32 continuously checks the current time and compares it with stored medicine schedules. When a match is detected, the system activates the buzzer and the corresponding tray LED, and displays the medicine information on the screen. User input through buttons is monitored to determine whether the medicine is taken or missed, and the corresponding status is updated in Firebase. In parallel, the system periodically reads health parameters such as heart rate, SpO_2, and temperature and uploads these values to Firebase for remote monitoring. The emergency button is continuously monitored, and upon activation, an alert is sent to Firebase with the corresponding timestamp.

Overall, the developed code ensures coordinated operation of all system components, enabling real-time reminders, health monitoring, emergency alert handling, and reliable communication with the cloud.

App Code Logic :

The mobile application for the proposed system is developed using the Flutter framework, enabling cross-platform functionality. The application provides an interface for doctors to enter prescriptions and for family members to monitor patient data remotely. The app uses firebase_core and firebase_database packages to communicate with Firebase Realtime Database. Data exchange is performed over the internet using the HTTP/HTTPS protocol, ensuring reliable and secure communication. The http package is used for uploading medicine images to Cloudinary.

The application includes modules for patient details, doctor details, and prescription management. Doctors can enter medicine names, timings, and instructions, which are stored in Firebase. The latest prescription is marked as active so that it can be accessed by the ESP32 device.
The family interface allows users to view prescription history, health parameters, and emergency alerts. The application retrieves real-time data from Firebase and displays medication status (taken/missed) and sensor readings.

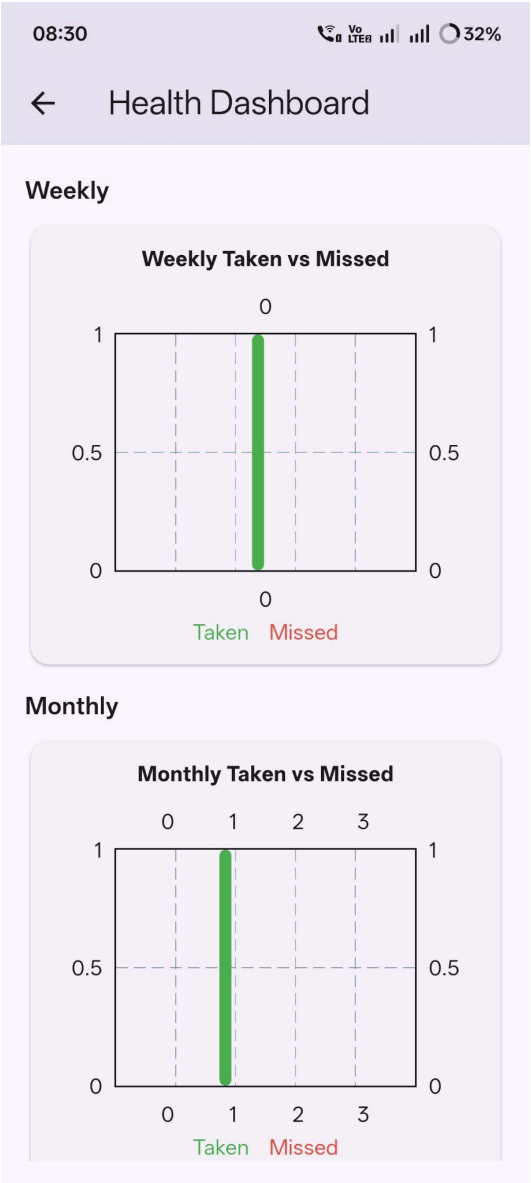
The overall working of the application is based on continuous synchronization with Firebase, ensuring that updates from both the device and the app are reflected in real time. This enables efficient prescription management and remote monitoring of the patient.

Results and discussion :

The IoT-based health monitoring and medication adherence system was successfully designed and tested. The ESP32 was able to connect to WiFi and communicate with Firebase for sending and receiving data. The medicine reminder feature worked properly by comparing the current time with the medicine schedule stored in Firebase. At the scheduled time, the buzzer was activated and the respective tray LED turned ON to indicate the correct medicine. The OLED display showed the medicine details, which helped the user identify the medicine easily. The Taken and Missed buttons were also tested, and the selected status was updated in Firebase with the timestamp.

The health monitoring part of the system was also tested using the MAX30102 and DHT11 sensors. The system was able to measure heart rate, SpO_2, and temperature values and upload them to Firebase. These values could be viewed through the mobile application by the doctor or family member. The emergency button feature also worked as expected by sending an alert to Firebase when pressed. During testing, some small issues such as WiFi delay, SSL connection errors, and sensor reading variations were observed, but these were handled using NTP time synchronization, proper WiFi handling, and validation conditions in the code. Overall, the system performed the expected functions and proved to be useful for medicine reminders, health monitoring, and emergency support.

Some photographs of the implemented system are shown below,



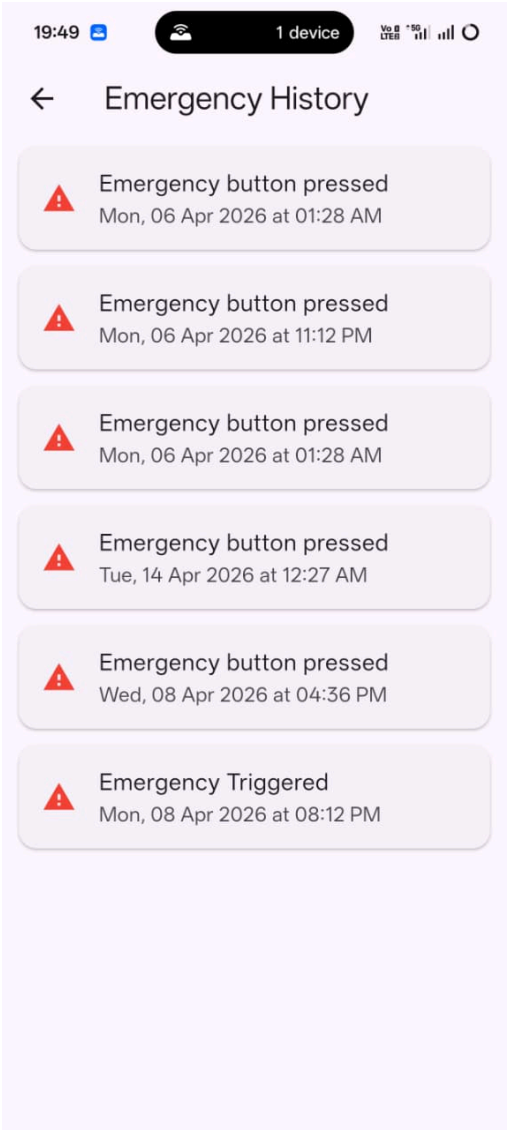


Fig 1 : Medication adherence result showing medicine taken status

Fig 2 : Emergency Button Pressed History

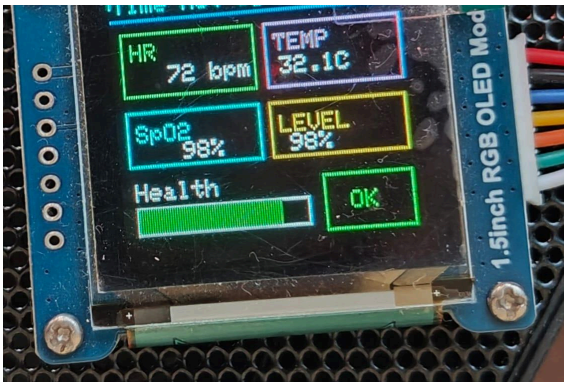


Fig 3 : Prescription details shown on display

Fig 4 : Sensor readings

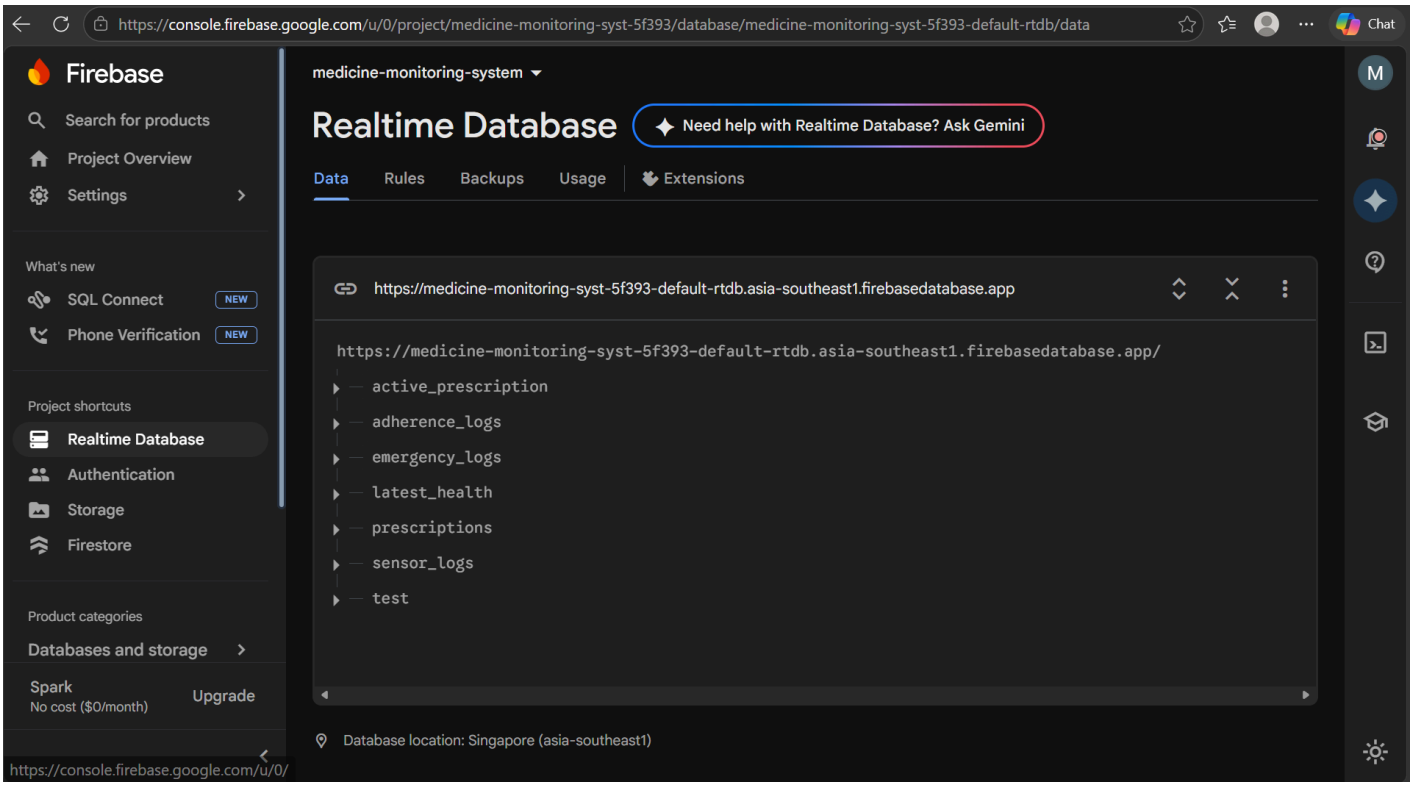


Fig 5 : Firebase cloud - used for storing the data

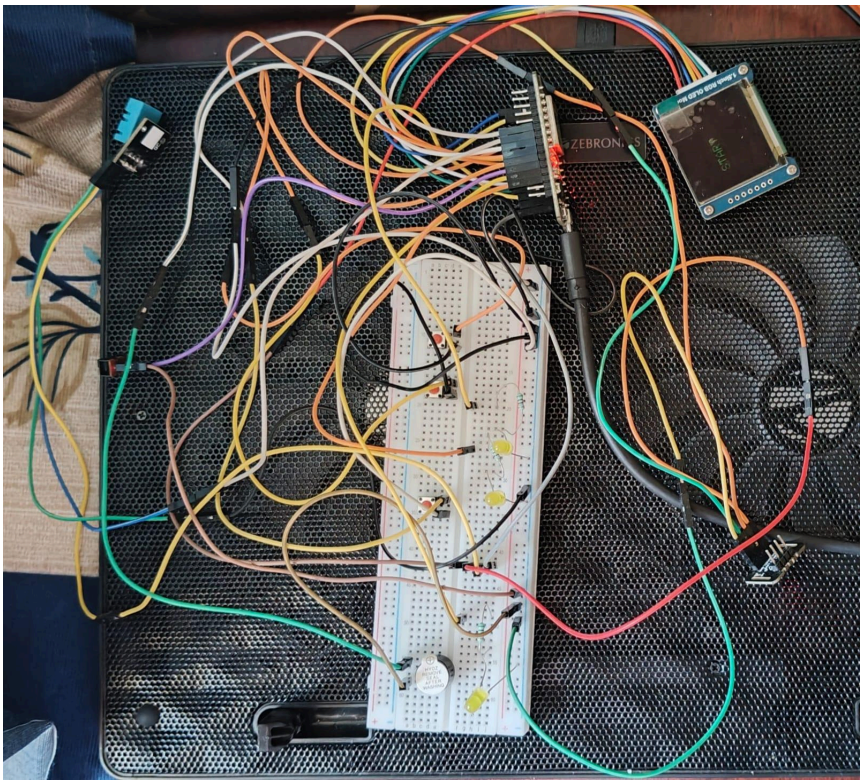


Fig 6 : Overall system Image

*(**Note:** In Fig. 4, the values shown are average readings. Since the sensors used in this project are basic sensors commonly used for general project work, they may not always provide highly accurate medical-grade results. In this case, the readings were close to the expected values; however, slight errors or variations can be observed during the video demonstration.)*

Conclusion :

By doing this project, we successfully developed an IoT-based health monitoring and medication adherence system using ESP32, Firebase, sensors, display, buzzer, LEDs and a Flutter mobile application. The system helps elderly patients take medicines on time by giving reminders and indicating the correct medicine tray. It also records whether the medicine is taken or missed and updates the information to Firebase for remote monitoring. Along with this, the system monitors basic health parameters such as heart rate, SpO_2 and temperature and makes the data available to family members or doctors through the mobile application.

This project shows how IoT can be used in healthcare to improve patient safety, especially for elderly people in rural areas. The emergency button feature makes the system more useful because the patient can send an alert during critical situations with a single button press. Although the system depends on internet connectivity and sensor accuracy, it provides a practical and low-cost solution for medication management and remote health monitoring. In the future, the system can be improved by adding GSM communication, mobile push notifications, more accurate biomedical sensors, and automatic medicine dispensing.

Summary :

The proposed IoT-based health monitoring and medication adherence system was successfully designed and implemented by integrating hardware, cloud services, and a mobile application. The system provides timely medication reminders using a buzzer and LED-based tray indication, along with real-time monitoring of health parameters such as heart rate, SpO_2, and temperature. The collected data is stored in Firebase and can be accessed through the mobile application, enabling remote monitoring by family members.

During the development of the system, several challenges were encountered. One of the major issues was unstable internet connectivity, which affected communication with Firebase. This was addressed by optimizing the frequency of data updates and ensuring proper WiFi connection handling. Another challenge was SSL connection errors with Firebase, which were resolved by implementing proper time synchronization using NTP before establishing secure communication. Additionally, sensor-related issues such as inaccurate readings and “no finger detected” conditions in the MAX30102 sensor were observed. These were handled by adding proper conditions in the code to avoid incorrect data processing during invalid readings. Integration of multiple components, including sensors, display, and cloud communication, also increased system complexity, which was managed by structuring the code efficiently and separating different functionalities. Overall, the system demonstrates a reliable and practical solution for improving medication adherence and providing real-time health monitoring. The challenges faced during development were successfully

resolved, resulting in a stable and functional system suitable for real-world applications, especially in rural environments.

USER GUIDE :

The developed system is designed to be simple and easy to use, especially for elderly users. The following steps describe how to operate the system:

1. **Power ON the Device:**
   Connect the device to a power supply. The ESP32 will automatically start and connect to the configured WiFi network.
2. **Initialization:**
   After powering ON, the system initializes all components, synchronizes time using NTP, and retrieves prescription data from Firebase.
3. **Prescription Entry (Mobile App):**
   The doctor enters patient details, medicines, timings, and instructions using the mobile application. The data is stored in Firebase and becomes available to the device.
4. **Medicine Reminder:**
   At the scheduled time, the system activates a buzzer and turns ON the LED corresponding to the specific medicine tray. The display shows the medicine details.
5. **User Response:**
   The user presses:
   - **TAKEN button** if the medicine is taken
   - **MISSED button** if the medicine is missed
     The status is updated in Firebase.
6. **Health Monitoring:**
   The system periodically measures heart rate, SpO_2, and temperature, and uploads the data to Firebase. These values can be viewed in the mobile application.
7. **Emergency Alert:**
   In case of an emergency, the user can press the emergency button. The system sends an alert to Firebase, allowing family members to respond immediately.
8. **Normal Display Mode:**
   When no reminder is active, the system displays patient information and health data on the screen.

Overall, the system requires minimal user interaction and is designed to assist elderly individuals in managing their medication and health effectively.

App user guide for both the doctor and family member. The mobile application is designed to be user-friendly and supports two roles: **Doctor** and **Family Member**. The working of each role is described below.

A. Doctor User Guide

1. **Login:**
   Open the application and select the **Doctor** role.
2. **Enter Patient Details:**
   Fill in patient information such as name, age, and gender.
3. **Enter Doctor Details:**
   Provide doctor name, hospital details, and diagnosis.
4. **Add Medicines:**
   Enter medicine details including name, type (tablet/syrup), quantity, and timings.
   Multiple medicines and multiple timings can be added.
5. **Add Instructions:**
   Select instructions such as before food, after food, or specific timing conditions.
6. **Upload Medicine Image (Optional):**
   Capture or upload an image of the medicine for easy identification.
7. **Save Prescription:**
   Save the prescription, which is uploaded to Firebase and set as the active prescription.
8. **View Preview and History:**
   The doctor can preview the prescription and view previously saved prescriptions.

B. Family Member User Guide

1. **Login:**
   Open the application and select the **Family Member** role.
2. **View Prescriptions:**
   Access the latest prescription and view details such as medicines, timings, and instructions.
3. **View Health Data:**
   Monitor real-time health parameters including heart rate, SpO_2, and temperature.
4. **View Medication Status:**
   Check whether medicines are taken or missed based on updates from the device.
5. **View Emergency Alerts:**
   Access emergency logs to see alerts generated by the patient along with date and time.
6. **Monitor Patient Remotely:**
   The application continuously retrieves updated data from Firebase, allowing remote monitoring.

Overall, the mobile application provides an efficient interface for managing prescriptions, monitoring health data, and ensuring patient safety through real-time updates and alerts.

Future Work :

- To increase the number of medicine trays for handling multiple prescriptions.
- To implement mobile push notifications for real-time alerts to family members.
- To improve accuracy by using advanced biomedical sensors.
- To add a voice assistant feature for hands-free emergency alerts.
- To implement AI-based analysis for predicting health conditions.
- To integrate the system with hospitals or healthcare providers.
- To enhance system reliability and reduce dependency on internet connectivity.
- To add GSM-based communication for areas with poor internet connectivity.
- To develop automatic medicine dispensing mechanisms.
- To improve the user interface of the mobile application for better usability.

Acknowledgements :

We would like to express our sincere gratitude to our project guide, **Prof. Bidhan Pramanick** sir, for his valuable guidance, support and encouragement throughout the development of this project. His suggestions, especially the idea of implementing the tray-based indication feature, and his support in providing the necessary components were very helpful in completing this work successfully. We also extend our sincere thanks to our lab assistant, **Parmeswar** sir, for his support in developing the device and for helping us understand the working of sensors and hardware components. We would also like to thank the Indian Institute of Technology Goa for providing the necessary resources and facilities to carry out this project.

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[2] Flutter Documentation, "Flutter Development Framework," Available: https://docs.flutter.dev

[3] Espressif Systems, "ESP32 Technical Reference Manual," Available: https://www.espressif.com

[4] Maxim Integrated, "MAX30102 Pulse Oximeter and Heart-Rate Sensor Datasheet," Available: https://datasheets.maximintegrated.com

[5] Arduino, "Arduino IDE Documentation," Available: https://www.arduino.cc

[6] Adafruit Industries, "SSD1351 OLED Display Library," Available: https://learn.adafruit.com

[7] The mobile app was developed using Flutter and we referred to standard Flutter tutorials and documentation for implementing UI, Firebase integration and real-time data handling. Also we took help of our CSE friends for logic and AI Tools for optimizing the code. Some links which we have referred to.

How to Install Flutter on Windows 11 | Android Studio

https://www.youtube.com/watch?v=TSIhiZ5jRB0

https://www.youtube.com/watch?v=VPvVD8t02U8

https://www.youtube.com/watch?v=5lDJNFSWUD8

[8]Google Drive link for all project resources, including developed codes and related files: IOT Project